

Design and Optimization of Piezoresistive MEMS Pressure Sensors using Abaqus and SEmulator3D



course -Materials, Design, and Manufacturing of Integrated Circuits, sensors, and MEMS (ME525)

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INTRODUCTION

Overview

Micro-Electro-Mechanical Systems (MEMS) pressure sensors are miniaturized devices that **convert applied pressure into an electrical signal**. Due to their compact size, low power consumption, and high sensitivity, MEMS pressure sensors are widely used in industrial, biomedical, automotive, and aerospace systems.

Application

- Transimission fluid sensing
- Gasoline direct injection
- Medium duty hydroulics

MEMS pressure sensors are widely used in industrial, biomedical, automotive, and aerospace systems.

PROBLEM STATEMENT

- **The Optimization Challenge:** Finding the ideal **diaphragm geometry** and **thickness** to requires balancing two opposing engineering requirements:
 1. Maximizing structural deflection to achieve peak sensor **sensitivity**.
 2. Limiting peak **Von Mises stress** so that it remains well below the fracture yield strength of single-crystal silicon under maximum loads.
- **Variables under study:** Diaphragm surface area (Square vs. Circular), Thickness variation (20, 25, 30, 35 μm), and overpressure conditions (2 to 4MPa).

OBJECTIVES

- **Geometric Modeling:** Design identical-surface-area square and circular diaphragms.
- **Finite Element Analysis:** Map stress concentrations and compliance curves using Abaqus.
- **Sensitivity Evaluation:** Extract deflection and structural sensitivity parameters to establish the superior geometry.
- **Virtual Fabrication:** Outline a cleanroom mask integration workflow using SEmulator3D.

SENSOR STRUCTURE

- **Diaphragm:** Thin silicon layer acting as the mechanical transducer.
- **Piezoresistors:** Strategically placed at areas of maximum stress concentration to capture maximum resistance change ($\Delta R/R$).

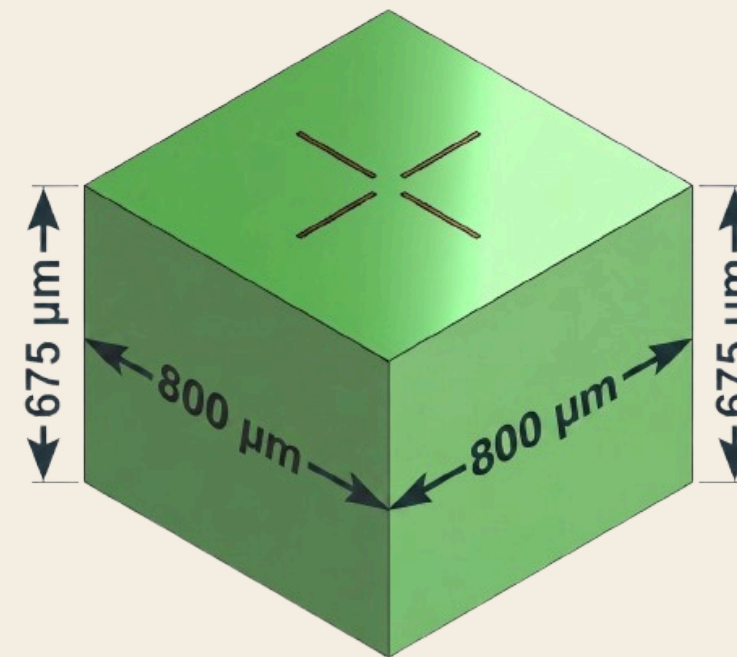


Fig 1: Diaphragm

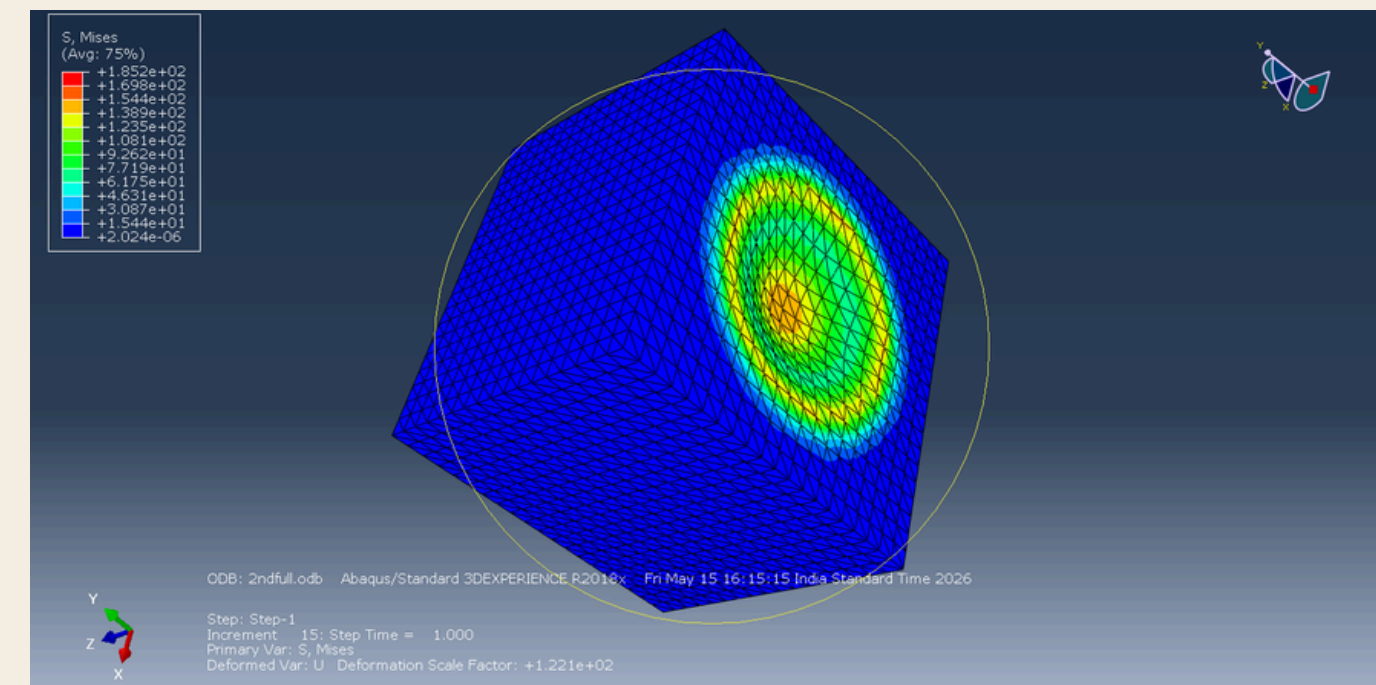
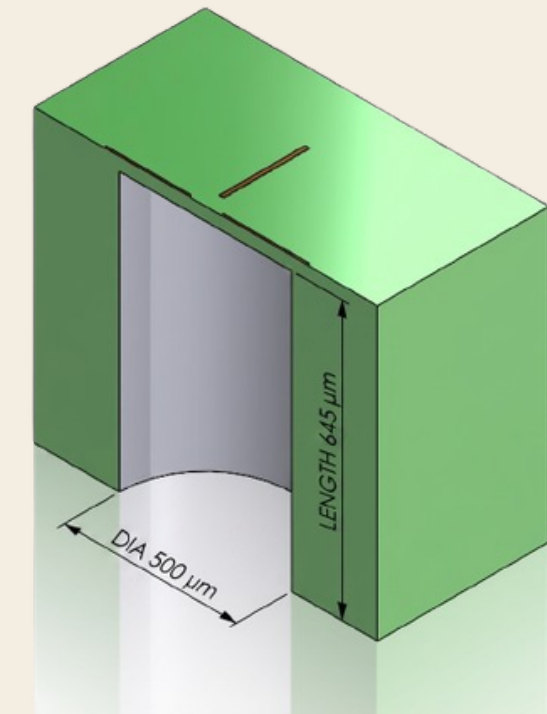


Fig 2: FEA of Diaphragm

SENSOR STRUCTURE & COMPONENTS

Wheatstone Bridge: Connects 4 active piezoresistors in a bridge configuration to translate Delta R into a clean differential output voltage (V_{out}).

WHEATSTONE BRIDGE CIRCUIT

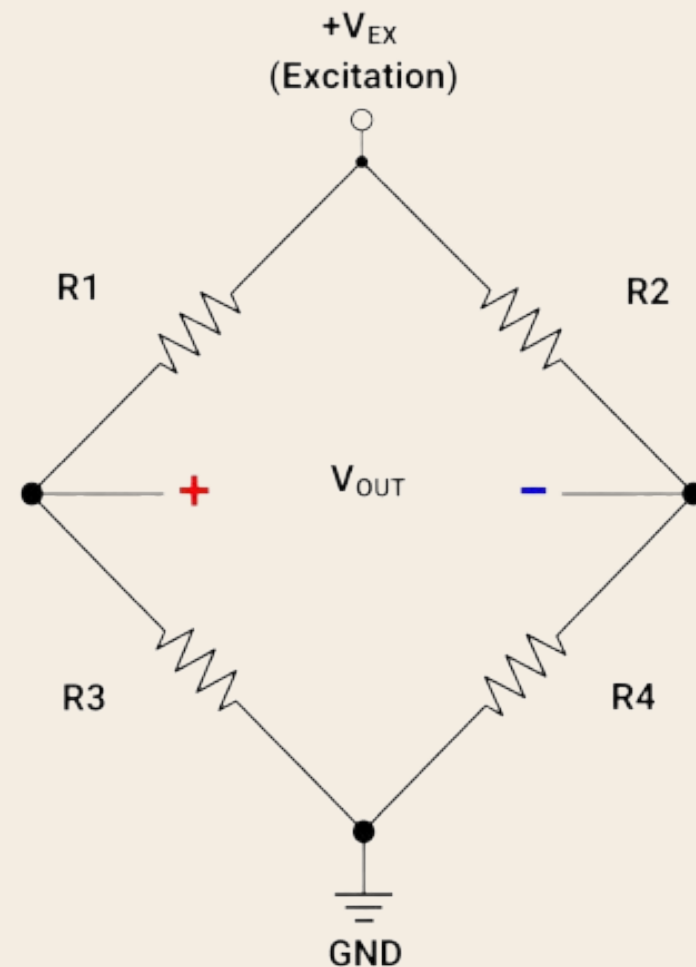


Fig 3: Wheatstone Bridge

BRIDGE OUTPUT EQUATION

$$V_{OUT} = V_{EX} \times \left(\frac{R_3}{R_1 + R_3} - \frac{R_4}{R_2 + R_4} \right)$$

For small ΔR (full bridge):

$$V_{OUT} \approx \left(\frac{\Delta R}{R} \right) \times V_{EX}$$

COMBINED EQUATION

$$V_{OUT} = \left(\frac{\Delta R}{R} \right) V_{EX} = (\pi \sigma) V_{EX}$$

PIEZORESISTIVE RELATION

$$\frac{\Delta R}{R} = \pi \sigma$$

π = Piezoresistive coefficient

σ = Stress in the resistor

METHODOLOGY

- **Software Workflow:** Models are built and meshed using solid continuum elements in Abaqus/CAE.
- **Material Model:** Single-Crystal Silicon, Elastic Modulus $E=200\text{GPa}$, Poisson's Ratio=0.28.
- **Element Shape:** Hexahedral (Hex) mesh
- **Element Type:** C3D8R
- **Global Element Size:** 0.02 mm

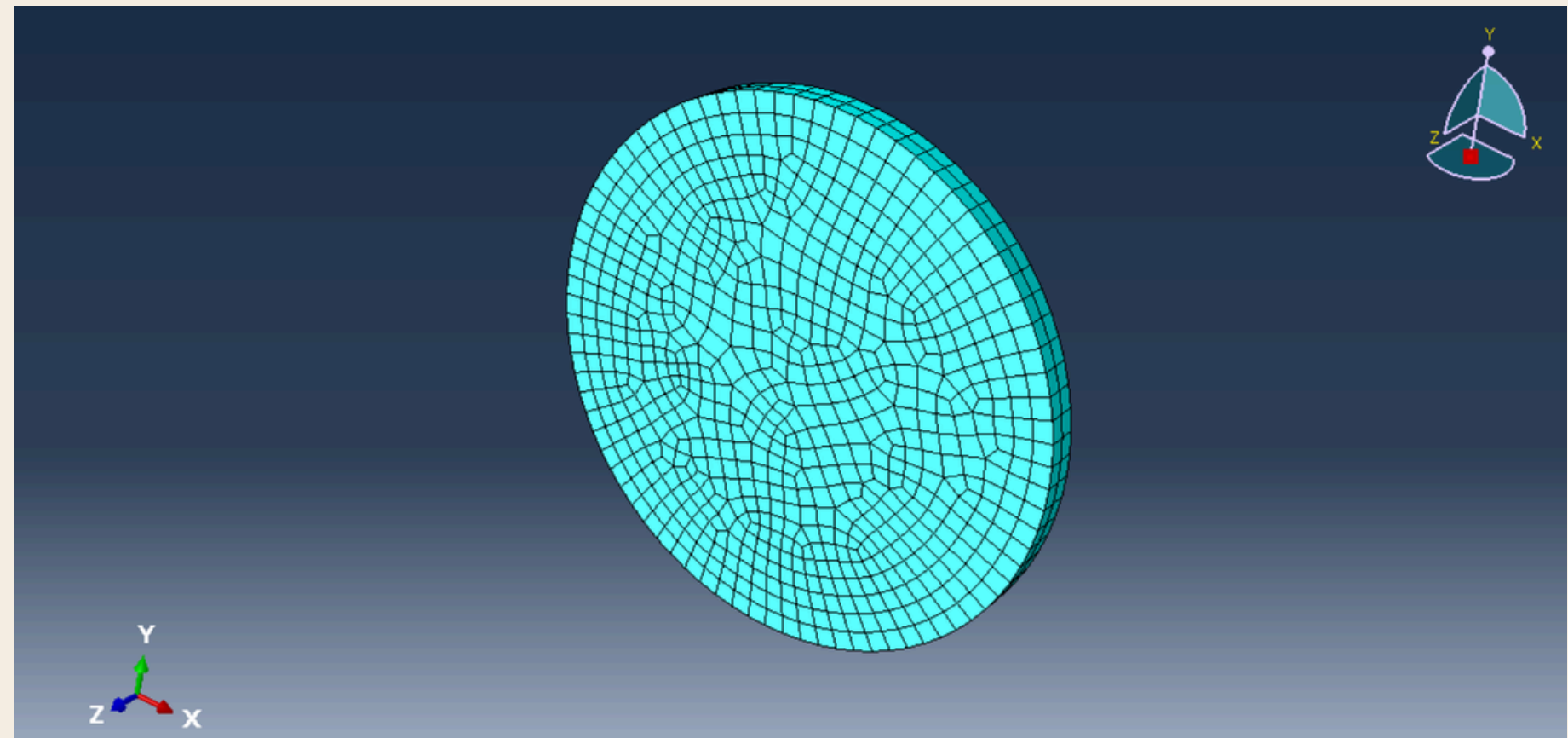


Fig 4: Meshing in ABAQUS

METHODOLOGY

- **Boundary Conditions:** The diaphragm edges were modeled using fully fixed ($U_x = U_y = U_z = U_{R_x} = U_{R_y} = U_{R_z} = 0$) (Encastre) boundary conditions.
- **Loading:** Uniform pressure distribution (2MPa and 4MPa) applied normal to the face.

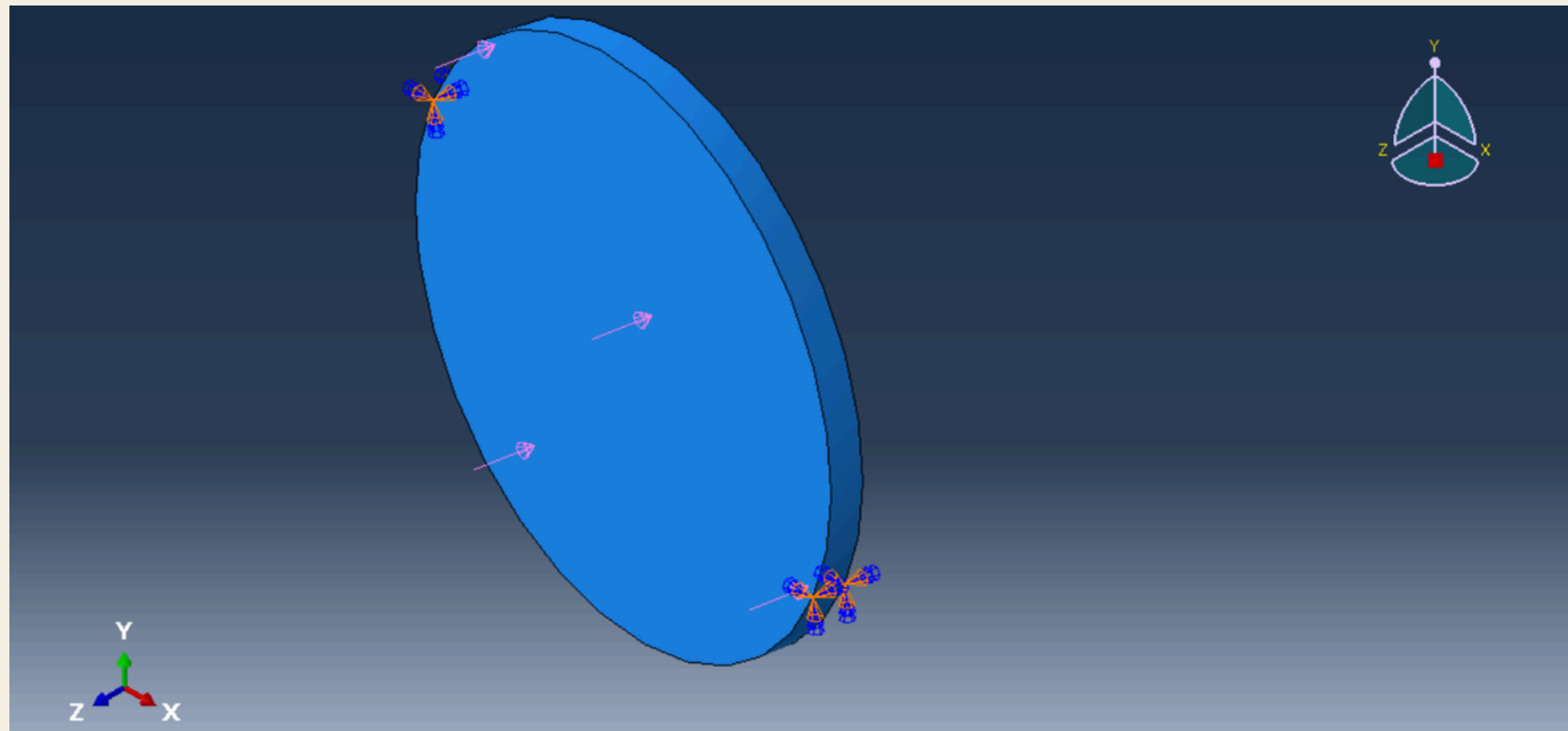


Fig 5: Boundary Conditions in ABAQUS

DESIGN AND OPTIMIZATION

- **Objective:** To identify the optimal diaphragm geometry that maximizes sensor sensitivity and minimum stress while maintaining structural safety under pressure loads.
- The diaphragm shapes that have been simulated FEM software diaphragm are such that the **area is same** in both case.

Diaphragm type	Dimensions (μm)
Circular	250(Radius)
Square	443(side)

DESIGN AND OPTIMIZATION

Max stress and max deflection values from Abaqus for different diaphragm thicknesses(20, 25, 30 & 35) across both shapes.

Square Shape			
Thickness (μm)	Applied pressure (MPa)	max Stress (MPa)	Max Deflection (mm)
20	2	162.6	0.000918
20	4	325.2	0.001836
25	2	111.6	0.0004063
25	4	224	0.00075
30	2	77.18	0.00024
30	4	154.4	0.0004801
35	2	61.31	0.0001475
35	4	117	0.000275

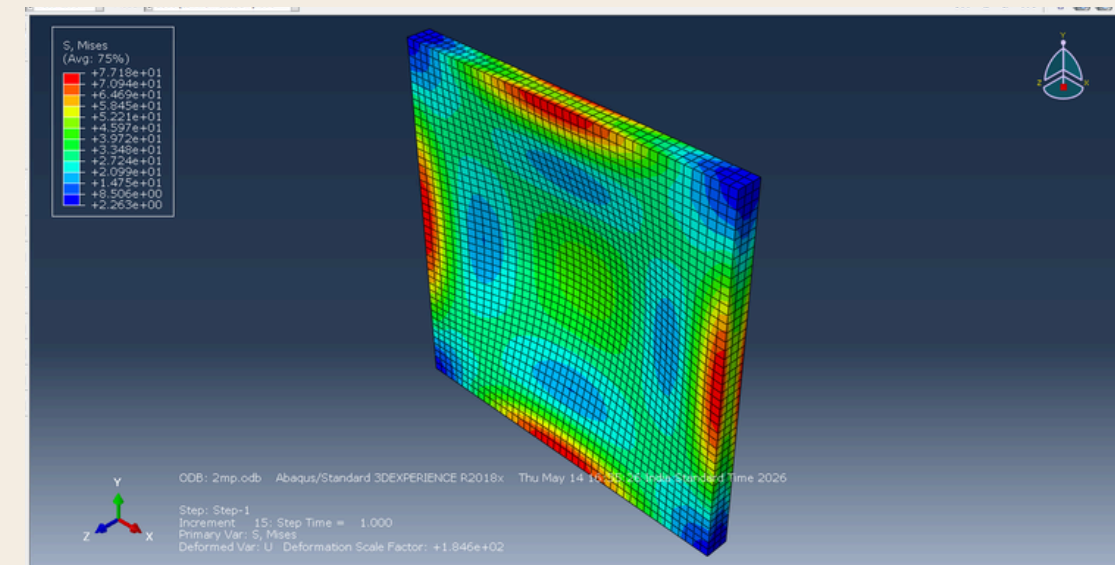


Fig 6: Stress Distribution in Square Diaphragm

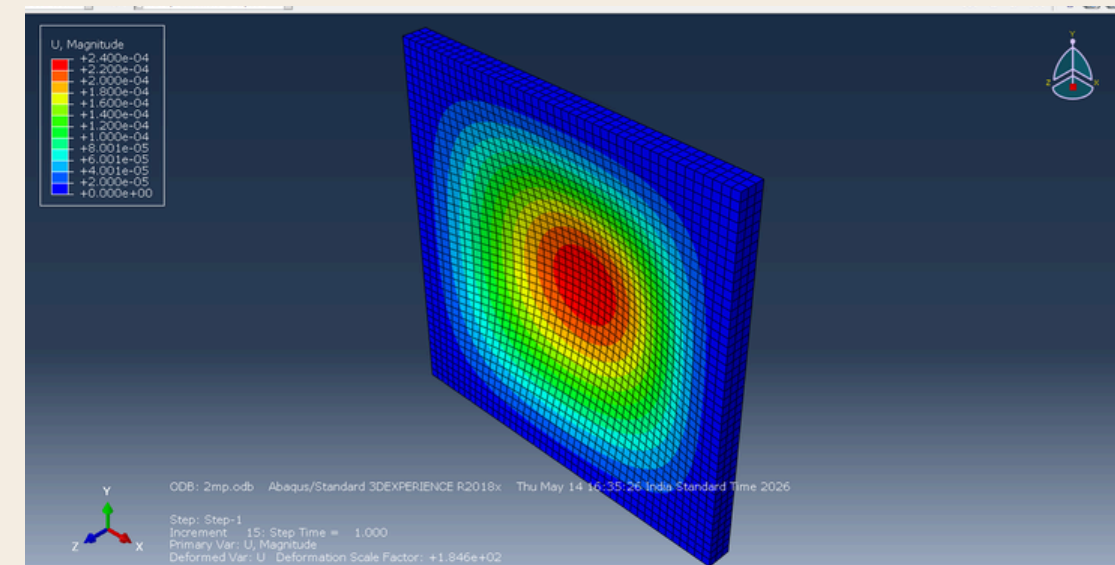


Fig 7: Deflection Distribution in Square Diaphragm

DESIGN AND OPTIMIZATION

Circular Shape				
Thickness (μm)	Applied pressure (MPa)	max Stress (MPa)	Max Deflection (mm)	
20	2	133	0.00114	
20	4	266	0.00228	
25	2	84	0.000502	
25	4	187	0.001	
30	2	66.2	0.000295	
30	4	133	0.00059	
35	2	50.3	0.00018	
35	4	101	0.00036	

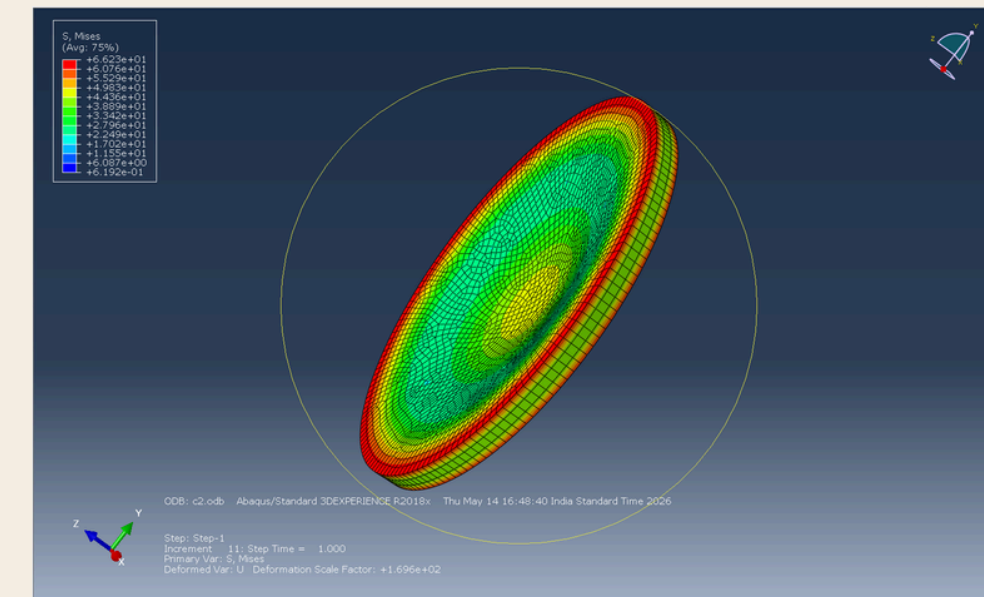


Fig 8: Stress Distribution in Circular Diaphragm

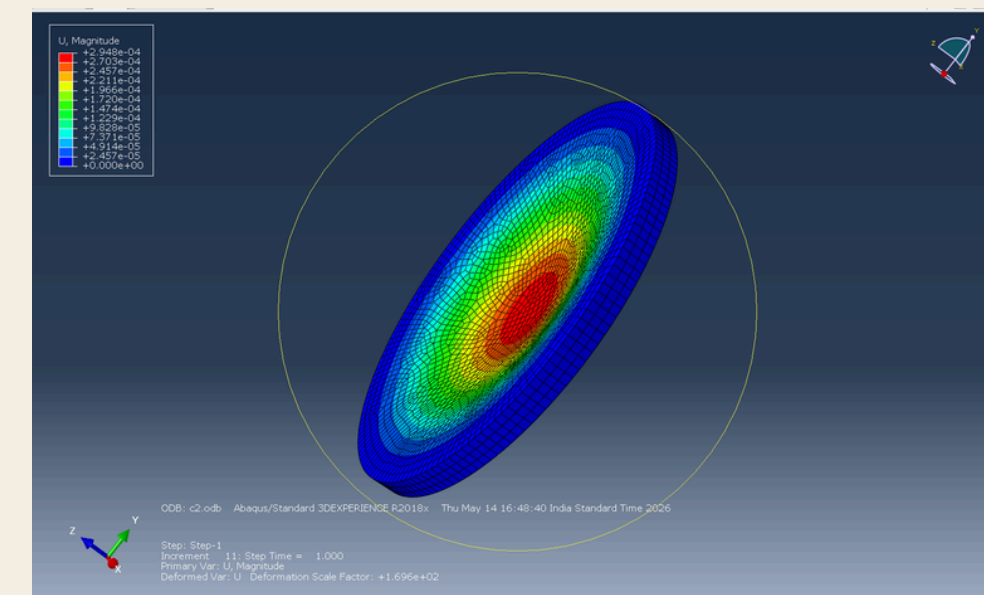


Fig 9: Deflection Distribution in Circular Diaphragm

DESIGN AND OPTIMIZATION

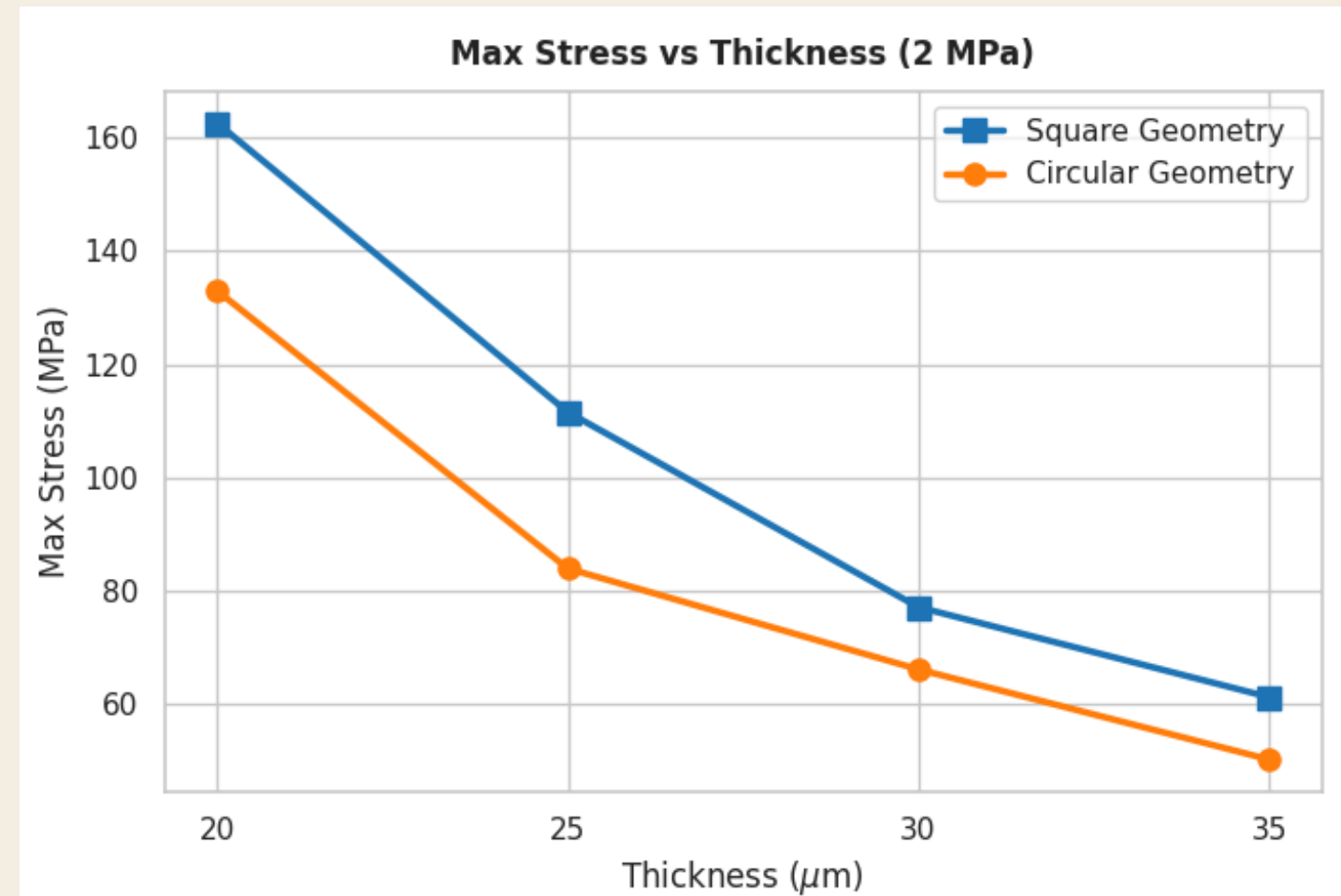


Fig 10: Max Stress Vs Thickness for 2MPa

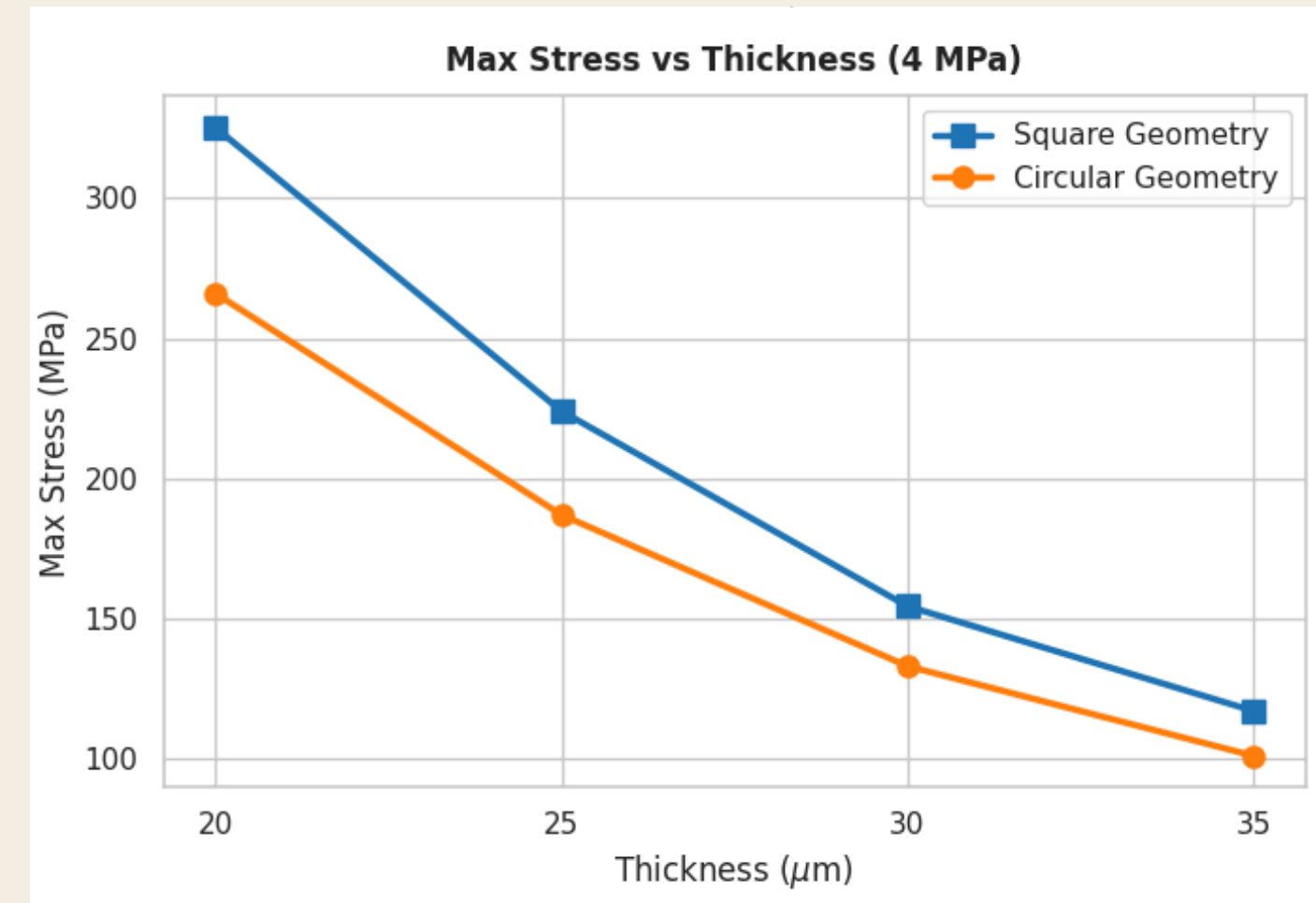


Fig 11: Max Stress Vs Thickness for 4MPa

Under all thickness conditions, the maximum stress in the circular diaphragm is consistently lower than that of a square diaphragm with the same surface area.

DESIGN AND OPTIMIZATION

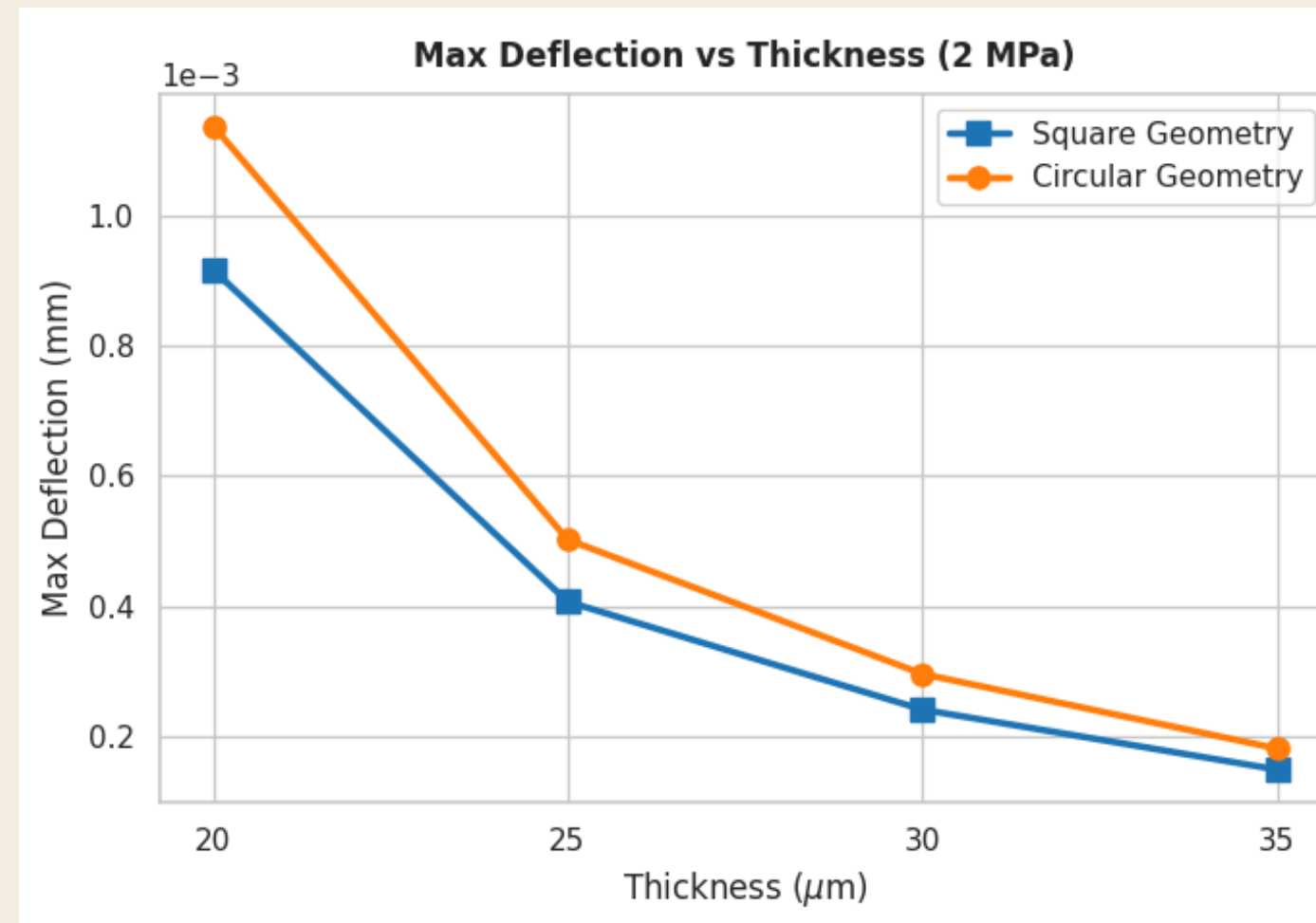


Fig 12: Max Deflection Vs Thickness for 2MPa

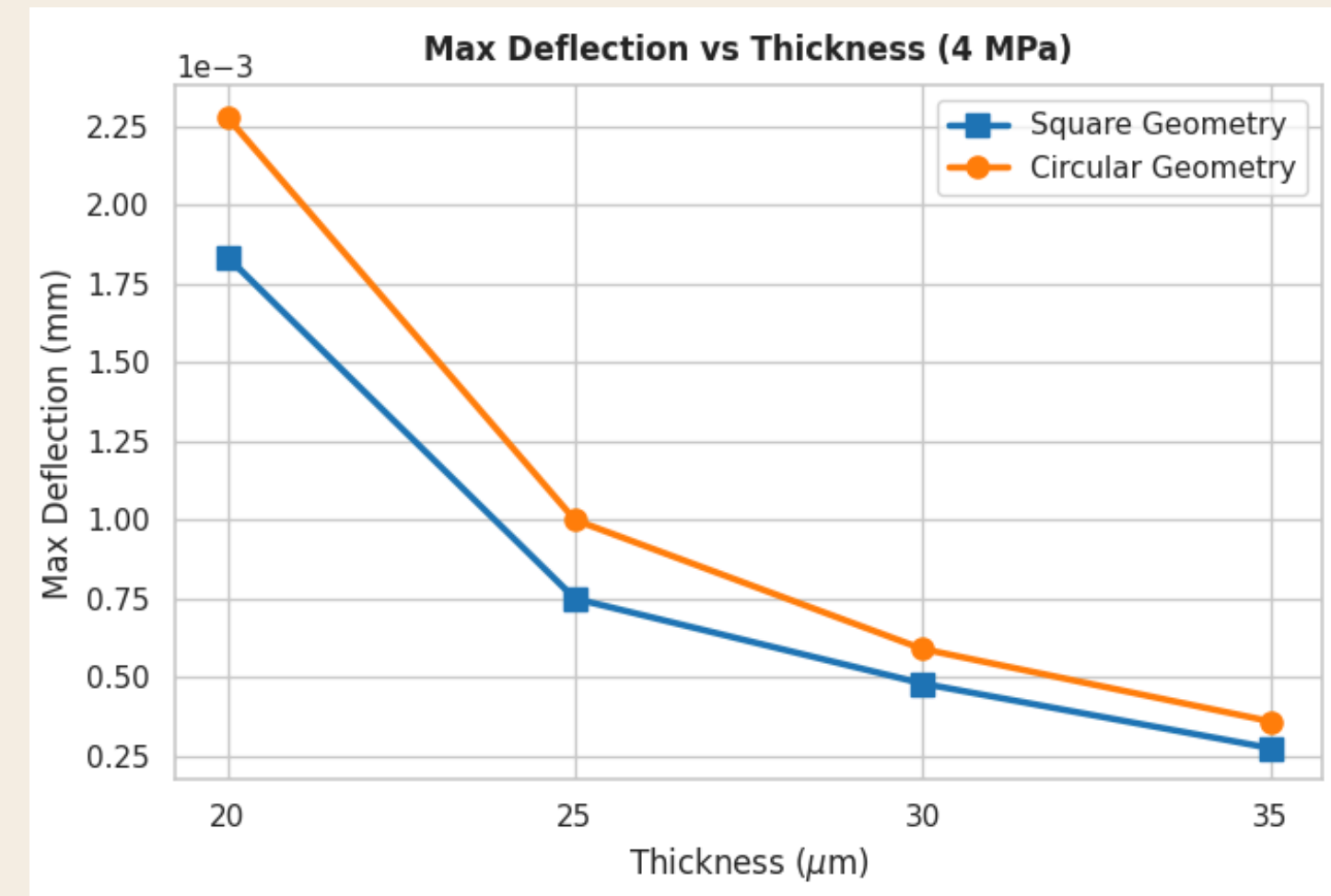


Fig 13: Max Deflection Vs Thickness for 4MPa

Under all thickness conditions, the maximum deflection in the circular diaphragm is consistently higher than that of a square diaphragm with the same surface area.

DESIGN AND OPTIMIZATION

A comparative analysis of **sensitivity** for circular and square diaphragm geometries over a thickness range of 20 to 35 μm at applied pressures of 2MPa and 4MPa.

Sensitivity (S) = $\frac{\text{Deflection } (\delta)}{\text{Applied Pressure (P)}} = \frac{\delta}{P}$

$S = \delta / P$

Thick- ness (μm)	Applied Pressure (MPa)	Square Max Deflectio n (mm)	Circular Max Deflection (mm)	Circular Sensitivity (mm/MPa)	Square Sensitivity (mm/MPa)
20	2	0.000918	0.00114	0.00057	0.000459
20	4	0.00184	0.00228	0.00057	0.000459
25	2	0.000406	0.0005017	0.00025085	0.00020315
25	4	0.00075	0.001003	0.00025075	0.0001875
30	2	0.00024	0.000295	0.0001474	0.00012
30	4	0.00048	0.00059	0.000147425	0.000120025
35	2	0.000148	0.00018	0.00009	0.00007375
35	4	0.000275	0.00036	0.00009	0.00006875

DESIGN AND OPTIMIZATION

A comparative analysis of sensitivity for circular and square diaphragm geometries over a thickness range of 20 micron to 35 micron at applied pressures of 2MPa and 4MPa.

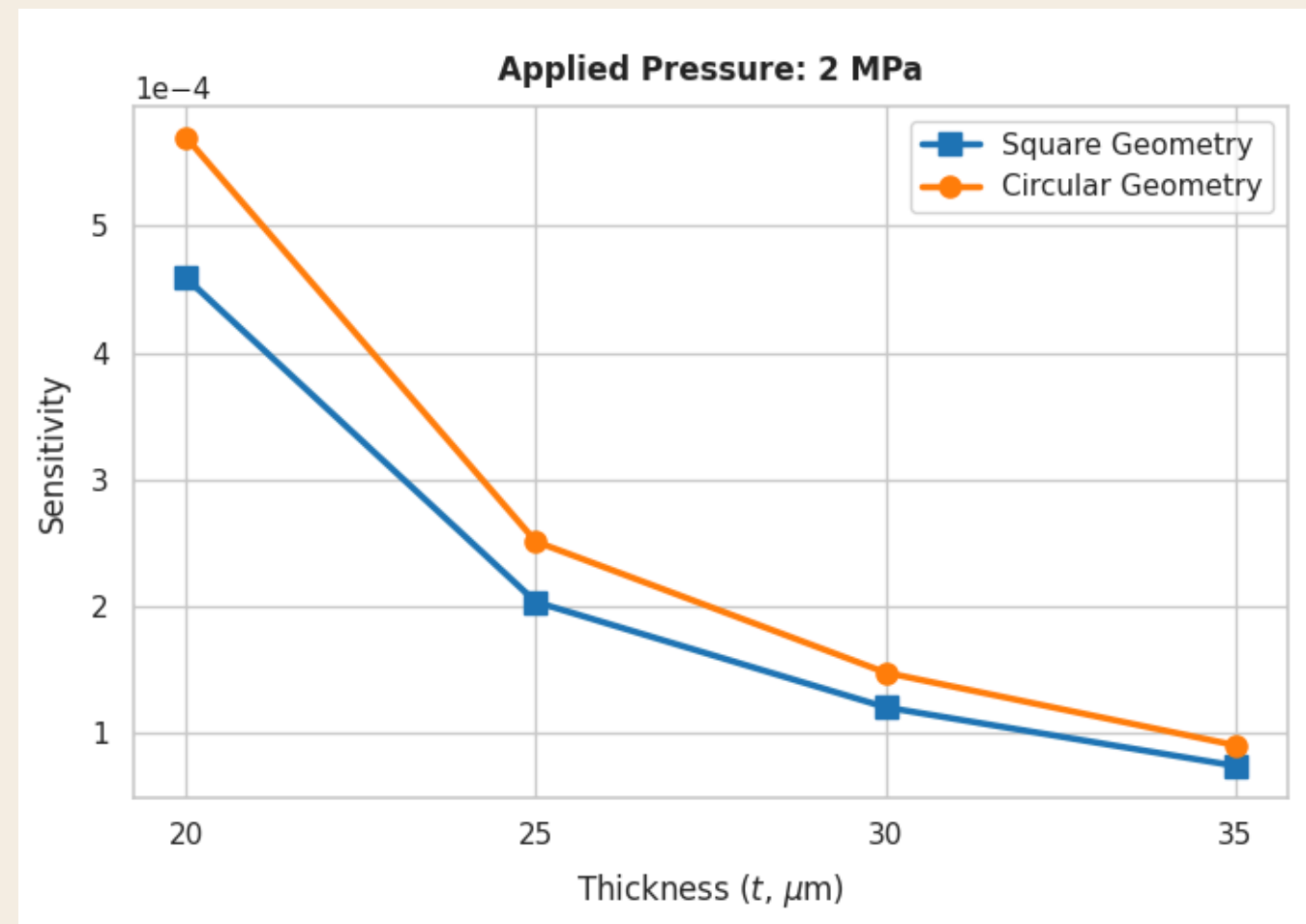


Fig 14: Sensitivity Vs Thickness for 2MPa

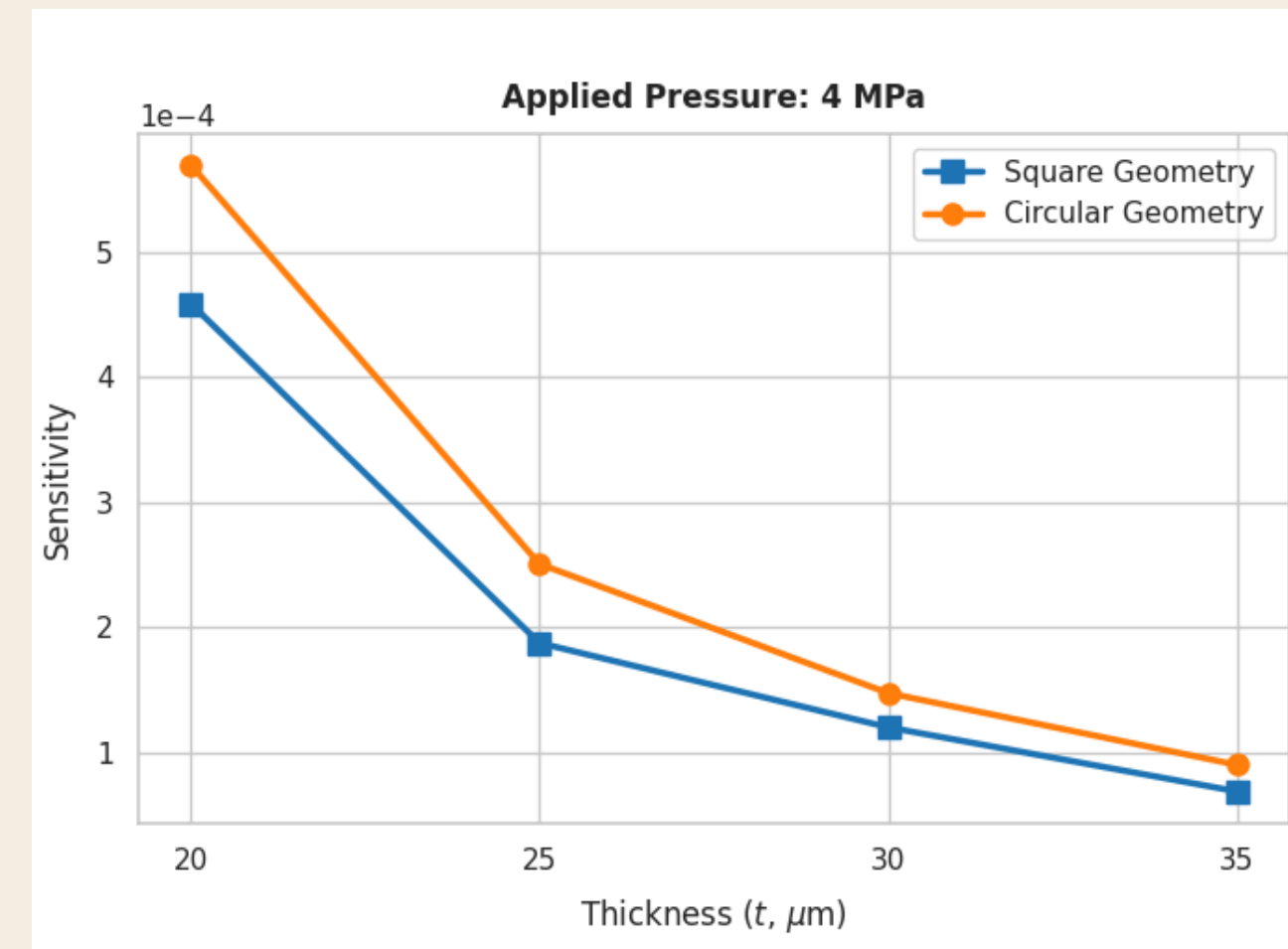


Fig 15: Sensitivity Vs Thickness for 4MPa

The circular diaphragm shows higher sensitivity compared to the square diaphragm.

RESULTS AND DISCUSSION

Mechanical Analysis using Abaqus

- Stress concentration was observed near the **clamped edges** of the diaphragm.
- Maximum deflection occurred at the **center** of the diaphragm.
- As diaphragm thickness increased, both stress and deflection decreased.

Geometry Comparison

- Circular diaphragm showed more **uniform stress distribution**.
- The circular geometry has higher sensitivity compared to the square geometry.

Thickness Optimization

- 20 μm $\rightarrow$ Highest sensitivity, lower structural safety
- 35 μm $\rightarrow$ Higher structural strength, lower sensitivity
- 25–30 μm $\rightarrow$ **Optimum balance** between sensitivity and reliability

PROCESS FLOW & VIRTUAL FABRICATION

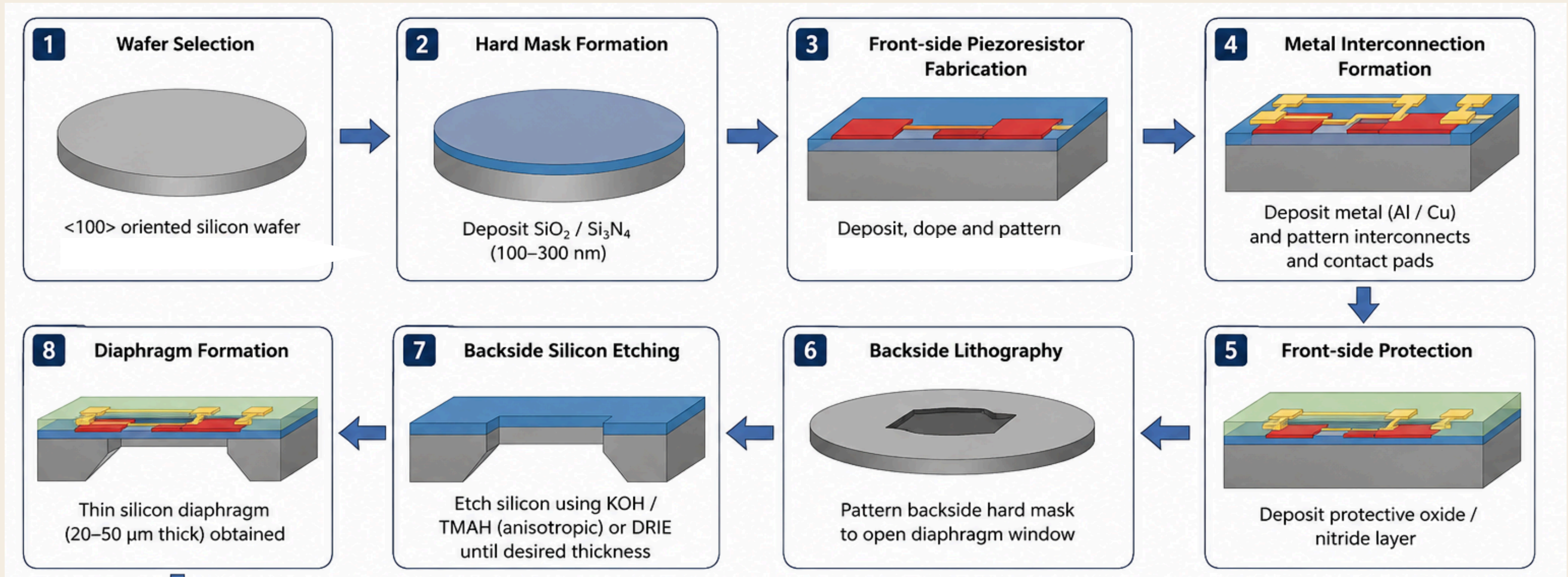


Fig 16: Steps of Fabrication

MASK DESIGN FOR FABRICATION

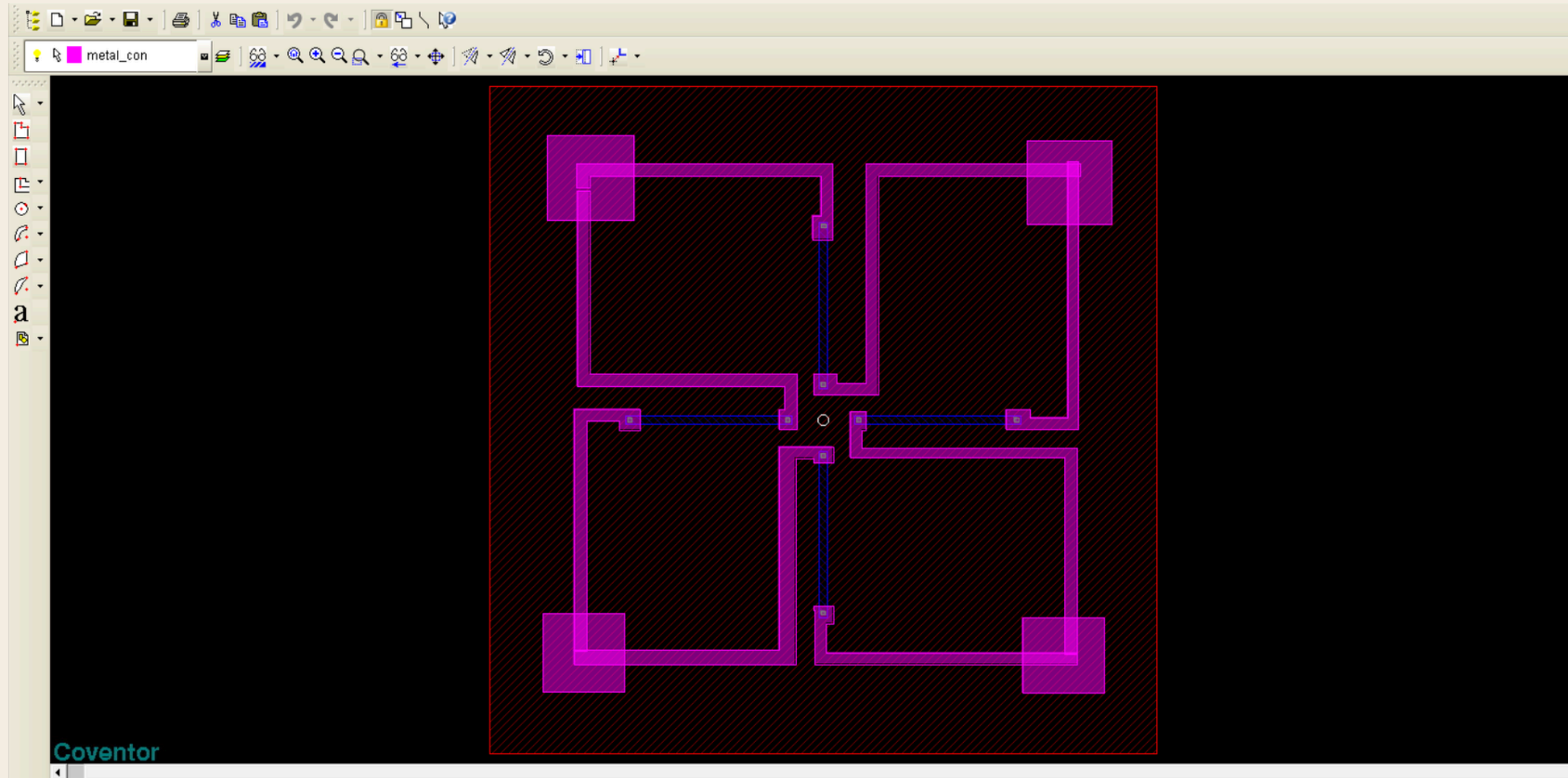


Fig 17: Mask

SEmulator3D Virtual Simulation Execution

Process Steps								
Number	Lock	Action	Name	Material(s)	Dopant(s)	Depth	Mask	Mask Polarity 
1		 Wafer Setup	Wafer Setup	/Silicon/Si_Xtal		30		
2		 Straight Dep...	Straight Deposit	/Oxides/SiO2_T...		1		
3		 Straight Dep...	Straight Deposit	/Organics/Resist		2		
4		 Expose Mate...	Expose Material	/Organics/Resist		3	Resistor	Dark
5		 Diffusion (A...	Diffusion (Requi...	/Oxides/SiO2_T...	/PType/B			
6		 Implant (Ad...	Implant (Requir...		/PType/B			
7		 Remove Mat...	Remove Materials	/Organics/Resist				
8		 Straight Dep...	Straight Deposit	/Oxides/SiO2_T...		2		
9		 Straight Dep...	Straight Deposit	/Organics/Resist		1		
10		 Expose Mate...	Expose Material	/Oxides/SiO2_T...		3	connector_dot	Dark
11		 Enhanced Ba...	Basic Etch	/Organics/Resis...		2		
12		 Planarizing ...	Planar Deposit	/Metals/Cu		1		
13		 Planarizing ...	Planar Deposit	/Metals/Cu		1		

Fig 18: Fabrication Steps in Semulator3D

CONCLUSION

- **Stress concentration** was observed near the clamped edges, while **maximum deflection** occurred at the **diaphragm center**.
- The **circular diaphragm** exhibited **lower stress concentration** and **higher deflection** compared to the square diaphragm.
- Sensitivity increased with decreasing diaphragm thickness, while structural strength improved with increasing thickness.
- The **25–30 μm** diaphragm thickness provided an optimum balance between **sensitivity** and **structural reliability**.
- Virtual fabrication and mask design workflow were successfully demonstrated using SEmulator3D.

REFERENCE

- (1). Design and Analysis of High-Performance and Small-Size Piezoresistive Pressure Sensors Chengwu Gao , Xuanqing Hua , and Dacheng Zhang
- (2). Design and Optimization of Piezoresistive MEMS Pressure Sensors Using ABAQUS Nabiollah Abol Fathi and Zohreh Allah Moradi
- (3). A High-Performance Micro Differential Pressure Sensor Xutao Fan 1, Lei Wang 2 and Songsong Zhang.
- (4). Design and Analysis of MEMS based Piezoresistive Pressure sensor for Sensitivity Enhancement * A. Nallathambia , T. Shanmugananthamb and D. Sindhanaiselvic.

THANK YOU